

Monolithically Integrated Quad-Line Comb Generation in Thin-Film LiNbO₃ enabling 3.2 Tb/s Superchannel Transmission

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Abstract We realize a 3.2 Tb/s superchannel (4x800 Gb/s/λ) at 1550 nm by using monolithic integrated quad-carrier generation. The fabricated TFLN device exhibits 36% on-chip conversion efficiency and 32 dB suppression of interjacent sidelines. Employing 81.1 GBd DP-64QAM, the BER after 100.8 km remains below 2E-2. ©2026 The Authors

Introduction

For coherent optical interfaces in the Tb/s regime, multi-carrier superchannel architectures have been proposed as an attractive way to circumvent electronic bandwidth limitations and the need for high-order modulation formats beyond 64 quadrature-amplitude modulation (QAM) [1,2].

Moreover, the pursuit of reducing size, power consumption, and cost drives integration of functions like optical sources, modulators, and multiplexers on photonic integrated circuits (PICs). Chip-scale comb-based multi-carrier generation have been reported mainly on the silicon nitride platform where microring resonator Kerr frequency combs have enabled transmission capacities from 1 up to beyond 50 Tb/s using tens to hundreds of tightly spaced carriers [2-4]. Lately, Thin-film lithium niobate (TFLN) has emerged as a highly attractive platform, combining a large Pockels coefficient with low loss, broad transparency, and compatibility with wafer-scale fabrication [5,6]. In this platform, electro-optic (EO) broadband comb generation in microring resonators was demonstrated [7].

However, Kerr combs as well as EO combs typically comprise tens to hundreds of spectral lines, resulting in a low optical power per generated comb line, rendering them less suited for use in superchannel transmitters where typically

only a few number of carriers are required.

In this work, we demonstrate a monolithically integrated quad-carrier comb generation on TFLN and use it to realize a 3.2 Tb/s (4 x 800 Gb/s/λ) superchannel transmission. Four power-balanced coherent carriers are generated from an external CW laser using an advanced EO microring architecture with an on-chip conversion efficiency of 36%. These four carriers are individually modulated with 81.1 GBd dual polarization (DP) 64QAM, yielding 3.89 Tb/s gross data rate.

We characterize the device and validate its suitability for superchannel transmission over 100.8 km single-mode fiber. At an optimum superchannel launch power of +11 dBm, all four sub-channels are received with bit-error ratios (BER) below the soft-decision forward error correction (SD-FEC) threshold [8].

Monolithic Integrated Quad-Carrier Generation

The experimental setup including the schematic and a photograph of the integrated electro-optic quad-carrier device is shown in Fig. 1. The PIC was fabricated using the TFLN multi-project wafer run at HyperLight [9]. It generates a power-flat four-carrier comb output spectrum from a continuous wave (CW) carrier, provided by an external cavity laser (ECL, 100 kHz linewidth) at 193.4 THz which is amplified to +20 dBm before

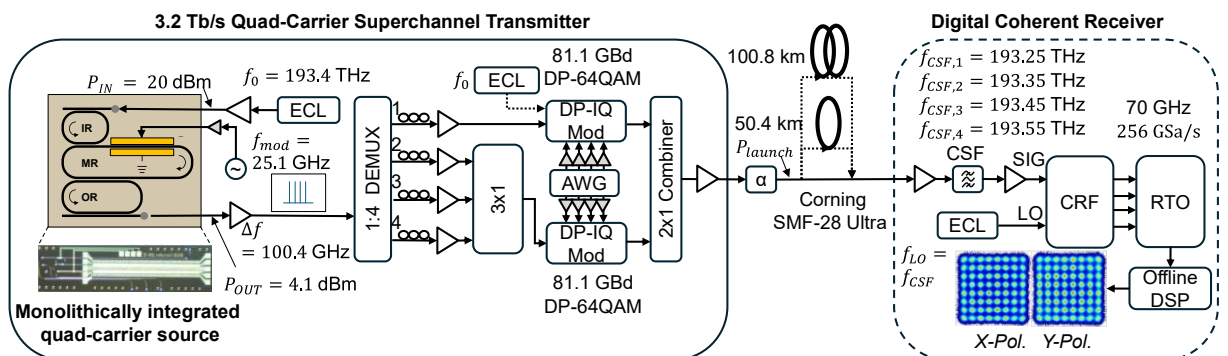


Fig. 1: Experimental setup. Photograph shows fabricated quad-carrier comb device PIC.

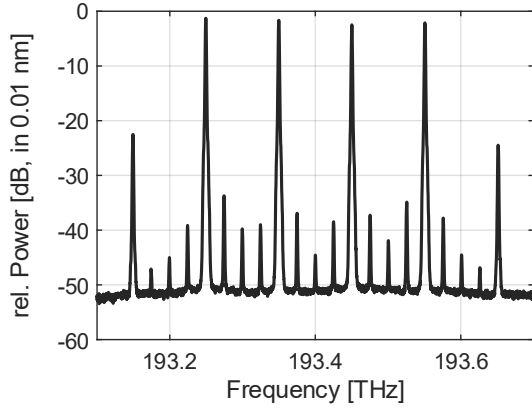


Fig. 2 Output quad-carrier spectrum

being fed into the PIC via a grating coupler. The waveguide layout consists of a main ring (MR) with a phase modulator section, driven at an RF frequency f_{mod} , surrounded by two ring resonators at the input (IR) and output (OR) for frequency-selective optical coupling. The IR and OR introduce spectral shaping. Additionally, the MR's free-spectral range is intentionally detuned by about 800 MHz from f_{mod} to limit the number of carriers generated.

The PIC is designed such that an electrical sinusoidal RF drive signal at $f_{\text{mod}} \approx 25$ GHz and 25.5 dBm power yields four dominant optical output carriers at a spacing of $\Delta f = 4f_{\text{mod}} \approx 100$ GHz, symmetrically arranged around the injected optical CW frequency. The finesse of the OR is designed to be about 100 to suppress interjacent side-lines at the output of the device before carrier extraction via a grating coupler.

The measured output spectrum of the PIC is shown in Fig. 2. It was measured after the output grating coupler using an optical spectrum analyzer (OSA) with a resolution bandwidth of 0.01 nm. We measure a total optical output power of +4.1 dBm. Considering the measured combined loss of input and output grating couplers of 11.5 dB, the on-chip power conversion efficiency is found to be 36.3%. The flatness of the four intended carrier lines is ± 0.6 dB. The interjacent sidelines next to the carriers are suppressed by >32.3 dB.

The spectrum also shows two additional (unintended) edge carriers which are suppressed by >20 dB but which fall outside of the subsequent carrier demultiplexer (see description of experimental setup).

Superchannel Transmission Setup

According to Fig. 1, the output of the quad-carrier comb device is amplified by an EDFA before being demultiplexed into four separate single-carrier signals. The four-port programmable optical demultiplexing filter (DEMUX, Coherent WaveShaper). is set to brick-wall bandpass

profiles around each carrier and provides ~ 40 dB suppression outside of the pass bands. The bandwidth is set to 80 GHz to deliberately include the adjacent sidelines and assess their impact on the performance of the superchannel. After demultiplexing, the carrier from DEMUX port 1, also referred to as channel under test (CUT), is fed to a dual-polarization in-phase and quadrature modulator (DP-IQ Mod.). The remaining three carriers from DEMUX ports 2, 3 and 4, acting as dummy channels, are combined using couplers (3x1) and fed to a second DP-IQ modulator. Individual per-carrier EDFAs are used to compensate for the losses of DEMUX and combiners while per-carrier polarization controllers are used to align each CW polarization to the DP-IQ modulators' inputs.

Both modulators are driven by the same arbitrary waveform generator (AWG) at 128 GSa/s to modulate the four carrier lines each with an 81.1 GBd DP-64QAM signal including 1.35% overhead for training sequences and having a root-raised cosine pulse shape with 0.01 roll-off. The AWG waveform includes nonlinear digital predistortion [10] tailored to the CUT, i.e., the upper DP-IQ modulator. The lower DP-IQ modulator (dummy channels) is connected to the inverted outputs of the AWG. Finally, the modulated optical sub-channels are combined to form a 3.2 Tb/s quad-carrier superchannel, consisting of one CUT and three dummy sub-channels. The dummy sub-channels serve two purposes: First, they emulate coherent crosstalk in the superchannel transmitter arising from finite rejection of the 1:4 DEMUX in conjunction with the passive combining of the four sub-channels. Second, they emulate nonlinear cross-channel distortions within the superchannel during transmission.

The quad-carrier superchannel is transmitted over 50.4 km and 100.8 km of Corning® SMF-28® Ultra fiber (loss ~ 9.6 dB per 50.4-km spool). The fiber launch power P_{launch} is adjusted using an EDFA (+17 dBm output power) followed by a variable optical attenuator (VOA, α). The transmit powers of all four sub-channels within the superchannel are matched using the four per-carrier EDFAs after the DEMUX. After transmission, an optically pre-amplified digital coherent receiver setup is used to recover the received signal. Following a cascade of two EDFAs with a channel selection filter (CSF, 95 GHz bandwidth) in between, the signal (SIG) is mixed with a CW local oscillator (LO) from an ECL in a DP coherent receiver frontend (CRF), whose electrical outputs are digitized using a 4-channel real-time oscilloscope (RTO, 256 GSa/s). The samples are processed using coherent data-aided offline digital signal processing (DSP) [11] including frontend

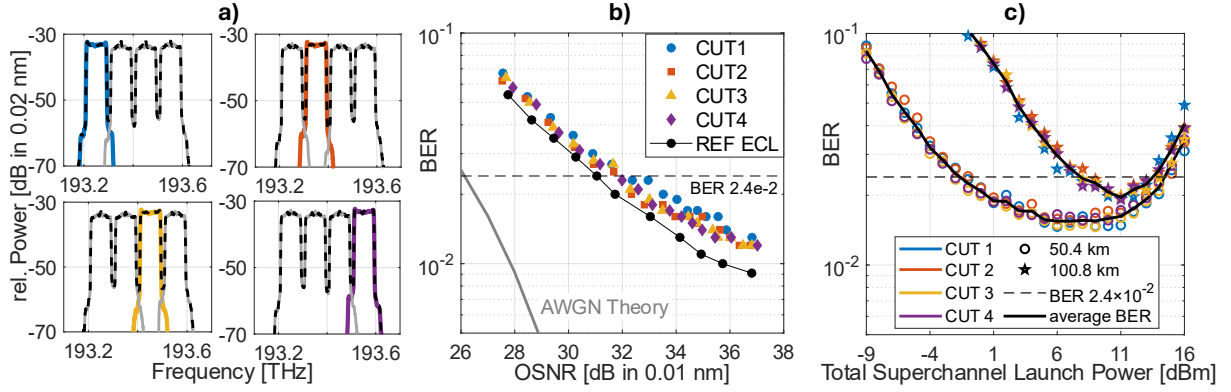


Fig. 3: (a) Optical spectra for all four superchannel configurations CUT1...CUT4: Modulated CUT (colored), dummy channels (grey) and superchannels (black), (b) Back-to-back BER performance, (c) Transmission performance over 50.4 & 100.8 km fiber

corrections, resampling to 2 samples per symbol, training-aided channel estimation and equalization, carrier phase recovery (blind phase search), 4x4 real valued DDLMS equalization [12], demapping and error counting.

The RTO as well as the balanced photodetectors inside the CRF had a limited electrical bandwidth of about 73.5 GHz. Therefore, the four sub-channels within the 400 GHz wide 3.2 Tb/s superchannel were received one after the other, by tuning the LO and the CSF to the respective carrier frequencies, while correspondingly swapping the filter profiles of the 1:4 DEMUX in the transmitter to always modulate the respective CUT using the upper DP-IQ modulator.

Results

The modulated spectra measured at the output of the 2x1 combiner are shown in Fig. 3(a). The dashed black lines show the quad-carrier superchannel spectra, while the colored and grey lines show the individual CUT and dummy sub-channel spectra, respectively. Each spectrum represents one out of four superchannel configurations (labelled "CUT1" to "CUT4") whereby the carrier frequency of the CUT is permuted across the four carriers from the quad-carrier generation.

Fig. 3(b) shows the back-to-back BER performance of the CUT in each superchannel configuration as a function of the per-subchannel optical signal-to-noise ratio (OSNR, 0.1 nm noise bandwidth). The OSNR was measured using an OSA before the channel section filter. We measured maximum OSNRs of about 37 dB and minimum BERs of about 1.2×10^{-3} for the CUTs in all four configurations. Compared to the reference performance (REF ECL) obtained when feeding the ECL directly into the upper DP-IQ modulator (cf. Fig. 1), the quad-carrier transmitter introduces about 1 dB OSNR penalty at the pre-FEC BER threshold of 2.4×10^{-2} for a soft-decision forward-error correction (SD-FEC) with 20% overhead [8]. This penalty is attributed to the

intentional inclusion of the adjacent sidelines within the DEMUX filter profiles and could be further reduced by tailored DEMUX filter profiles.

Finally, we analyzed the superchannel performance after fiber transmission over 50.4 km and 100.8 km. The BER of the CUT in each superchannel configuration as a function of the optical launch power of the superchannel into the transmission link is depicted in Fig. 3(c). We find an optimum superchannel launch power of about +7 dBm for 50.4 km transmission, yielding an average BER of the CUTs across all configurations of about 1.6×10^{-2} . At the increased transmission distance of 100.8 km and at an increased optimum launch power of +11 dBm we obtained an average BER of 1.9×10^{-2} while also all four individual sub-channel BERs were below the SD-FEC threshold.

Conclusion

We performed 3.2 Tb/s coherent superchannel transmission based on integrated quad-carrier generation on thin-film lithium niobate.

The fabricated PIC generated four optical carriers from a single external CW laser and exhibited an on-chip conversion efficiency of 36.3% and 32.3 dB sideline suppression. Using 81.1 GBd DP-64QAM modulation on each carrier yielded a net capacity of 3.2 Tbit/s (4×800 Gbit/s/λ) after accounting for 20% SD-FEC and 1.35% training overhead. After 100.8 km of standard single-mode fiber, all four sub-channels within the superchannel remained below the SD-FEC threshold, and the average BER was about 1.9×10^{-2} at a total superchannel launch power of +11 dBm.

These results demonstrate that few-line EO comb devices on TFLN are well suited as on-chip comb generation for coherent Tb/s superchannels. The demonstrated monolithic quad-carrier PIC, operated from a single CW laser, provides a concrete path towards scalable, co-integrated coherent superchannel transmitter PICs.

Acknowledgements

The SMF-28® Ultra Optical Fiber was kindly provided by Corning®.

The authors thank Shinsuke Tanaka (formerly with 1Finity Inc., Japan) for the valuable discussions.

This work was partly funded by the CELTIC-NEXT project SUSTAINET-Advance funded by the German Federal Ministry of Research, Technology and Space (BMFTR) under contract 16KIS2272.

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